6.1
Oxidation of Nitrogen and Phosphorus

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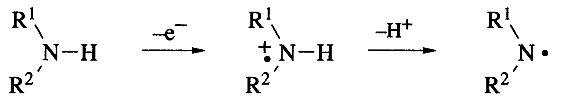
6.1.1 INTRODUCTION

The aim of this chapter is to describe the main types of functional group interconversion which involve oxidation at nitrogen or phosphorus. 'Oxidation' has been interpreted as including nitrosation, amination, halogenation and other reactions in which attack by an electrophilic heteroatom takes place. The material has been organized on the basis of the nature of the starting materials and the products, rather than on the mechanism of oxidation or the type of oxidant. For example, no attempt has been made to cover comprehensively the chemistry of aminium cation radicals[1] or of aminyl radicals[2] which can occur as intermediates in the oxidation of amines (Scheme 1). Also, oxidation which results in reaction at carbon, such as radical coupling through carbon or reaction at a C—H bond adjacent to nitrogen, is not covered in any detail here, even if the initial oxidation occurs at nitrogen. Reactions of C—H bonds activated by nitrogen are described in Chapter 2.5 of this volume. The biological oxidation of amines and other organic nitrogen compounds has also been reviewed elsewhere.[3]



Scheme 1

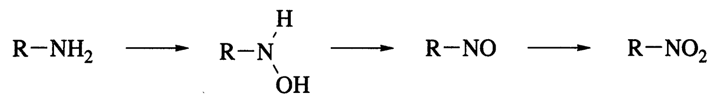
Of the many reviews on oxidation which are available, two in particular provide detailed coverage of the literature up to the early 1980s on aspects of oxidation at nitrogen. One of these, by Boyer, is a comprehensive survey of oxidation reactions of nitrogen compounds in which the number of oxygen atoms attached to nitrogen is increased.[4] The other, by Rosenblatt and Burrows, deals with oxidation of amines.[5] References to the primary literature have not always been included here if they are available from these two reviews.

6.1.2 OXIDATION OF THE NH_2 GROUP

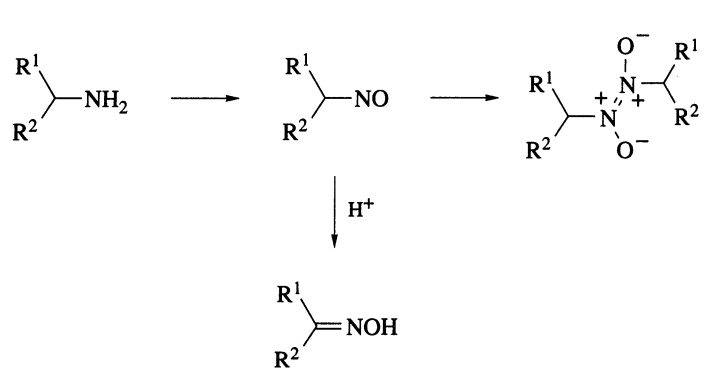
6.1.2.1 Primary Amines

6.1.2.1.1 Oxidation to hydroxylamines, nitroso compounds and nitro compounds

Aromatic and aliphatic primary amines can be oxidized to the corresponding nitro compounds by peroxy acids and by a number of other reagents.[5,6] The peroxy acid oxidations probably go by way of intermediate hydroxylamines and nitroso compounds (Scheme 2). Various side reactions can therefore take place, the nature of which depends upon the structure of the starting amine and the reaction conditions. For example, aromatic amines can give azoxy compounds by reaction of nitroso compounds with hydroxylamine intermediates; aliphatic amines can give nitroso dimers or oximes formed by acid-catalyzed rearrangement of the intermediate nitrosoalkanes (Scheme 3).



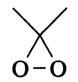
Scheme 2



Scheme 3

MCPBA has been regarded as the reagent of choice for the conversion of primary aliphatic amines into the corresponding nitro compounds.[5] The peroxy acid must be used in excess to minimize formation of dimers of the intermediate nitroso compounds. The yield of nitroalkane is also increased if the reaction is carried out at elevated temperature, since this favors the monomeric rather than the dimeric form of the intermediate nitrosoalkane and allows it to be oxidized further.[7] For example, cyclohexylamine gave the dimer of nitrosocyclohexane (43%) when oxidized by MCPBA at 23 °C, but at 83 °C (in boiling 1,2-dichloroethane) the only product was nitrocyclohexane (86%).

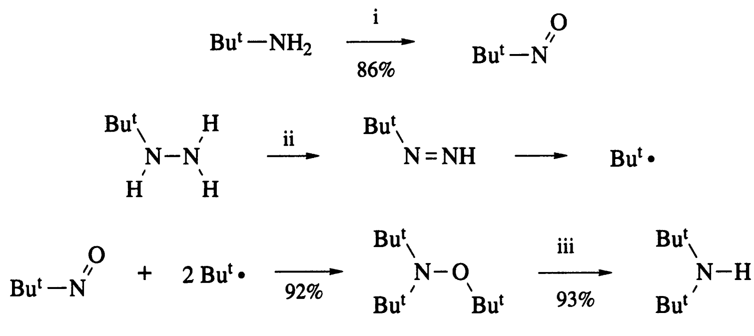
Ozone is an alternative oxidant for aliphatic primary amines.[8] The yields are generally not as good as with MCPBA, although a technique of 'dry ozonation' on silica at low temperature has been described which results in good conversions of cyclohexylamine and other aliphatic primary amines into the corresponding nitro compounds.[9] Solid sodium permanganate has been used to prepare 2-methyl-2-nitropropane from *t*-butylamine in good yield.[10] An oxidant which has given excellent yields of both aromatic and aliphatic nitro compounds from the corresponding amines is dimethyldioxirane (**1**).[11] The reagent is readily prepared *in situ* from acetone and a commercial oxidant, Oxone. When used under phase transfer conditions it is a mild, nonacidic oxidant which allows the selective oxidation of aromatic amines in the presence of indoles and furans. This reagent has also been used for the selective oxidation of primary amines to hydroxylamines.[12]



(1)

Aromatic primary amines have also been converted into the corresponding nitro compounds with peroxy acids: trifluoroperacetic acid, peroxymaleic acid and peracetic acid have all been used. A good alternative reagent for aromatic amines bearing electron-withdrawing substituents is sodium perborate in acetic acid. 4-Nitrobenzonitrile was prepared (91%) from 4-aminobenzonitrile with this reagent.[13] Hindered aromatic amines are oxidized only to the nitroso compounds: MCPBA[14] and perbenzoic acid[15] in stoichiometric amounts give the nitroso compounds in good yield. Thus, 2,6-difluoroaniline is oxidized to 2,6-difluoronitrosobenzene (85%) by perbenzoic acid, and other 2,6-disubstituted anilines react in a similar way. Fremy's salt can also oxidize hindered arylamines to the nitroso compounds.[4] Partial oxidation of *o*-phenylenediamine to 2-nitrosoaniline was achieved by dropwise addition of peracetic acid to the diamine.[4]

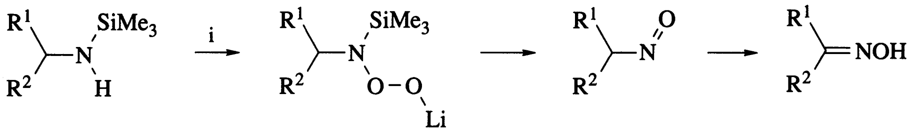
Corey and Gross have made use of the peracetic acid oxidation of *t*-butylamine and other tertiary amines to nitrosoalkanes in a synthesis of di-tertiary alkylamines.[16] This is illustrated in Scheme 4 by the synthesis of di-*t*-butylamine. Cyclohexylamine and other alkylamines have been oxidized to the nitroso compounds in high yield by a reagent consisting of sodium percarbonate, sodium hydrogen carbonate and tetraacetylethylenediamine in aqueous dichloromethane.[17] Several nitrosoalkanes have also been obtained from the corresponding amines by oxidation with aqueous hydrogen peroxide and sodium tungstate;[4] various other metal catalysts, including titanium(IV) and vanadium(V) species, have been used in combination with *t*-butyl hydroperoxide.[18] A procedure has been described for the low temperature oxidation of trimethylsilylamines to oximes, by way of the nitroso compounds (Scheme 5): the oxidant is



i, $MeCO_3H$, ii, PbO_2; iii, sodium naphthalenide

Scheme 4

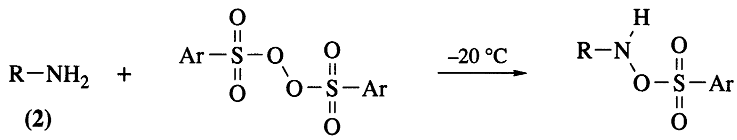
dry air, which reacts with the lithium salts of the amines.[19] This reaction is tolerant of phosphines, sulfides and other functional groups which are susceptible to oxidation.



i, BuLi, dry air, –60 to –40 °C

Scheme 5

Hydroxylamines are commonly postulated as intermediates in these oxidations, but they are rarely isolated or detected. Two examples of reactions in which products at the hydroxylamine oxidation level can be isolated both involve peroxides. Oxidation of alkylamines (**2**; R = Bu^t, Me and $PhCH_2$) with arenesulfonyl peroxides bearing an electron-withdrawing group gave the arenesulfonyloxyamines in good yields (Scheme 6).[20] A similar reaction of primary amines with dibenzoyl peroxide gave benzoyloxyamines.[21]

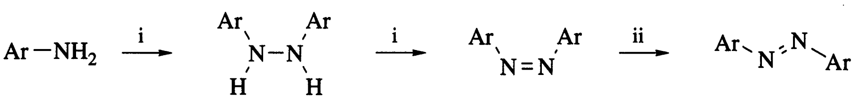


(2)

Scheme 6

6.1.2.1.2 Oxidation to azo compounds and related dehydrogenation

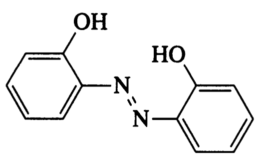
Aromatic azo compounds can be obtained by the oxidation of primary arylamines. The reagent most widely used for this purpose is activated manganese dioxide.[22] This converts aniline and substituted anilines into the corresponding azo compounds in moderate to good yield.[23] At room temperature the products are the *cis*-azobenzenes, which are isomerized to the *trans* compounds on heating.[24] It is probable that hydrazobenzenes are intermediates in the reaction, but these are more easily oxidized than the starting anilines (Scheme 7). These oxidations are inhibited by electron-withdrawing substituents and some nitro-substituted anilines fail to react.



i, MnO_2; ii, heat

Scheme 7

Nickel peroxide also oxidizes anilines to azoarenes.[25] Yields are moderate, although nitroanilines can also be oxidized with this reagent. Other reagents which have been used are silver oxide on Celite,[26] lead(IV) acetate[27] and barium manganate (which is claimed to give azo compounds in higher yield than manganese dioxide).[28] Bispyridinesilver permanganate, $py_2Ag^+MnO_4^-$, which is soluble in polar organic solvents, is also a good alternative to manganese dioxide.[29] Sodium perborate is a convenient oxidant for the conversion of *para*-substituted anilines into azo compounds.[30] Sodium hypochlorite has been used for the oxidation of pentachloroaniline and other chlorinated anilines,[31] while potassium superoxide is capable of selectively oxidizing *ortho*- and *para*-substituted diamines and aminophenols to the corresponding azo compounds in good yield.[32] Thus, 2-aminophenol gave the azo compound (**3**; 70%),



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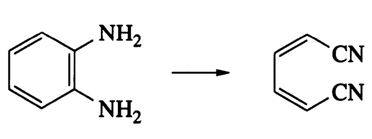
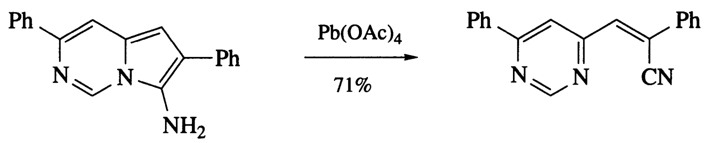


(4)

whereas *meta*-substituted anilines failed to react. Oxidation of the anilines to aminyl radicals, followed by their combination to hydrazobenzenes, can account for the observed selectivity.

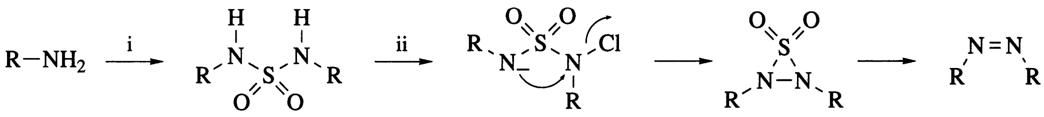
Manganese dioxide and potassium superoxide both oxidize *o*-phenylenediamine to the azo compound, but the more powerful oxidants nickel peroxide and lead(IV) acetate cause ring cleavage; the product, which can be isolated in low to moderate yield, is the (*Z*,*Z*)-dinitrile (**4**). This oxidative cleavage can be brought about in high yield with oxygen in the presence of copper(I) chloride and pyridine.[33] Some five-membered heteroaromatic amines can be cleaved to nitriles on oxidation: an example is the amine (**5**; Scheme 8) which undergoes ring opening on oxidation by lead(IV) acetate.[34]



(**5**)

Scheme 8

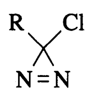
Nitriles are also the usual products of oxidation of aliphatic amines RCH_2NH_2 by nickel peroxide and lead(IV) acetate. Aliphatic azo compounds can be prepared from these primary amines by first converting them into sulfamides (**6**), these then being oxidized with sodium hypochlorite or (better) *t*-butyl hypochlorite (Scheme 9).[35] A few aliphatic azo compounds can be formed in good yield by direct oxidation of *t*-alkylamines; for example, AIBN was formed (86%) by oxidation of the amine $Me_2C(CN)NH_2$ with sodium hypochlorite. A special case of azoalkane formation is the synthesis of chlorodiazirines (**7**) from amidines $RC(=NH)NH_2$ by oxidation with sodium hypochlorite.[36]



(**6**)

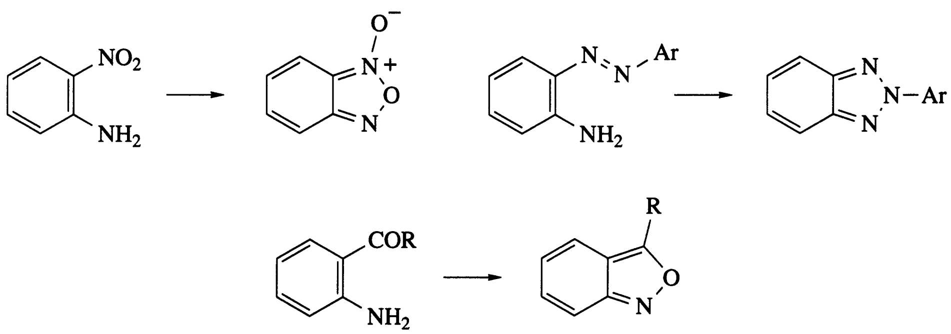
i, SO_2Cl_2, pyridine; ii, Bu^tOCl, Bu^tOK

Scheme 9



(**7**)

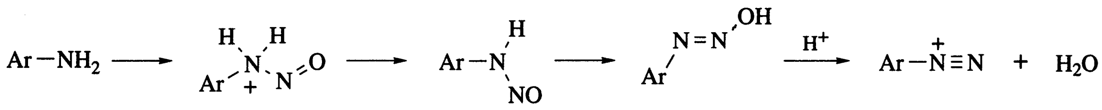
Oxidation of primary aromatic amines bearing a nucleophilic substituent at the *ortho* position can provide a useful route to some heterocyclic compounds. Some examples, shown in Scheme 10, are syntheses of benzofuroxans,[37] benzotriazoles[38] and benzisoxazoles.[39]



Scheme 10

6.1.2.1.3 Diazotization

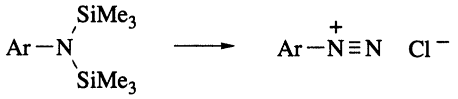
The formation of aromatic diazonium salts from aromatic primary amines is one of the oldest synthetic procedures in organic chemistry. Methods based on nitrosation of the amine with nitrous acid in aqueous solution are the best known, but there are variants which are of particular use with weakly basic amines and for the isolation of diazonium salts from nonaqueous media. General reviews include a book by Saunders and Allen[40] and a survey of preparative methods by Schank.[41] There are also reviews on the diazotization of heteroaromatic primary amines[42] and on the diazotization of weakly basic amines in strongly acidic media.[43] The diazotization process (Scheme 11) goes by way of a primary nitrosamine.



Scheme 11

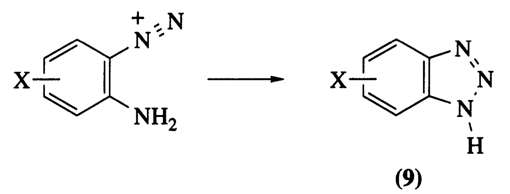
If the amine is basic enough to form a salt with dilute mineral acids in aqueous solution, the normal diazotization method of adding sodium nitrite in aqueous solution to a solution or suspension of the amine salt is satisfactory. Variations in the order of addition of the reagents are sometimes used, for example, when there is another functional group present which is sensitive to nitrous acid, or if the amine salt is very insoluble.[40] Weakly basic amines can often be diazotized successfully either in concentrated sulfuric acid or in a mixture of sulfuric acid with acetic or phosphoric acid.[43] It is likely that nitrosylsulfuric acid is formed on addition of sodium nitrite to concentrated sulfuric acid, and that this is the nitrosating agent when the amine is subsequently added.

Diazotization in organic solvents allows solid diazonium salts to be isolated. Diazotization can be carried out using an ester of nitrous acid, such as pentyl nitrite, in a solvent such as acetic acid or methanol. A procedure has also been described for isolating diazonium tetrafluoroborates, in excellent yield, by carrying out the diazotization with boron trifluoride etherate and *t*-butyl nitrite in ether or dichloromethane at low temperature.[44] Another method for the preparation of a variety of diazonium salts in a nonaqueous medium makes use of the chemistry of bis(trimethylsilyl)amines (**8**).[45] These compounds react in dichloromethane with nitrosyl chloride and other nitrosating agents which are generated *in situ*. Thus, benzenediazonium chloride was isolated (96%) from bis(trimethylsilyl)aniline.

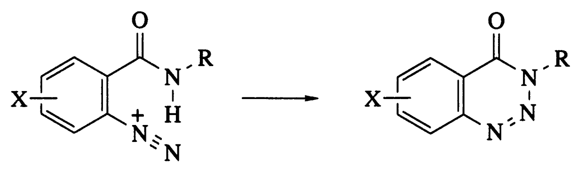


(8)

The diazotization of aromatic amines with a nucleophilic substituent at the *ortho* position is a common method of synthesis of benzo-fused heterocyclic compounds with two or more contiguous nitrogen atoms. Benzotriazoles (**9**), benzotriazinones (**10**), and benzothiadiazoles (**11**) are examples of heterocyclic ring systems that can be prepared in this way.



(9)



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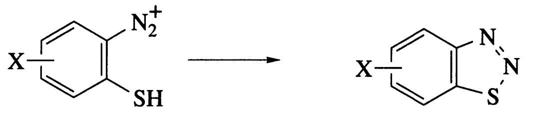
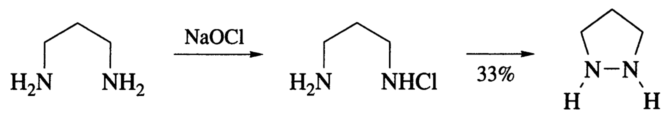


(11)

6.1.2.1.4 Amination

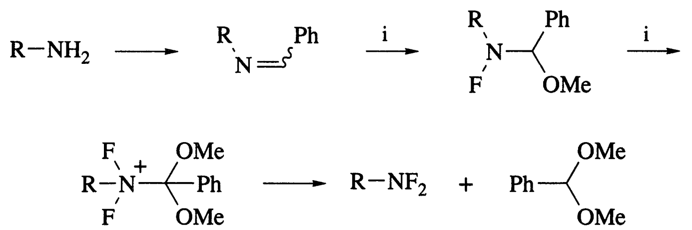
Simple primary alkylamines can be converted into the corresponding monoalkylhydrazines in moderate yield by amination with chloramine or with hydroxylamine-*O*-sulfonic acid.[46] The method is tolerant of the presence of double bonds: allylhydrazine was prepared (52%) by reaction of chloramine with allylamine. The method is not generally applicable to the preparation of 1,2-disubstituted hydrazines from primary alkylamines and *N*-chloroalkylamines although intramolecular examples are known.[47] Tetrahydropyrazole was prepared in moderate yield in this way (Scheme 12) and piperazine was prepared in low yield by the same type of reaction.



Scheme 12

6.1.2.1.5 Halogenation, sulfenylation and related reactions

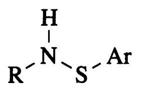
Methods for the *N*-chlorination and *N*-bromination of amines have been reviewed.[48] Alkylamines are chlorinated by aqueous sodium hypochlorite, chlorine in aqueous sodium bicarbonate, *N*-chlorosuccinimide, or *t*-butyl hypochlorite at low temperature. By using the appropriate amount of chlorinating agent, selective mono- and di-chlorination can be achieved. Although some of these compounds, especially those derived from *t*-alkylamines, are stable enough to be isolated, the majority are unstable since they can undergo further reaction associated with the loss of a proton from the α-carbon atom. *N*-Chloroanilines are also unstable unless the ring is substituted by an electron-withdrawing group, because of the tendency of the chlorine to migrate to a ring carbon atom. Reagents which have been used for *N*-bromination include bromine and aqueous sodium hypobromite. *N*-Haloamines have also been prepared by the action of the appropriate halogens on *N*-trimethylsilylamines. An indirect method of *N*,*N*-difluorination of *t*-alkylamines is illustrated in Scheme 13.[49]



i, CF_3OF, MeOH

Scheme 13

The reaction of primary amines with arenesulfenyl halides leads to the formation of sulfenamides (**12**).[50] These compounds are most stable when the aryl group has electron-withdrawing substituents at the 2- and 4-positions. Selenamides are formed in an analogous manner but are somewhat less stable: aliphatic amines give isolable compounds, but most anilines react with selenyl halides to give products of ring substitution.[51] An example of an isolable selenamide is compound (**13**), which was prepared (79%) from the amine and 2-nitrobenzeneselenyl chloride.[52] This compound was used as an intermediate in the preparation of 7α-methoxycephalosporins.



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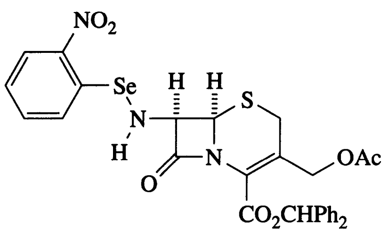


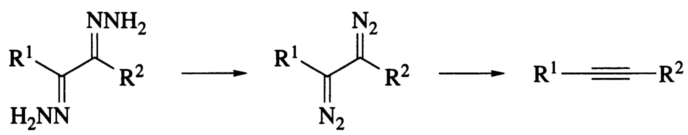
(13)

6.1.2.2 Hydrazones, Hydrazines and Hydroxylamines

6.1.2.2.1 Dehydrogenation

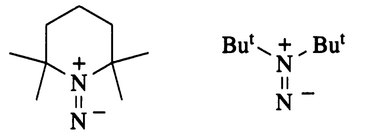
This section includes synthetically useful oxidative reactions of substrates of the general types $R^1R^2C{=}NNH_2$, $R^1R^2NH_2$ and $RONH_2$, where the groups R^1, R^2 and R can be alkyl, aryl or acyl.

The oxidation of many hydrazones provides a method of preparation of the corresponding diazo compounds $R^1R^2C{=}N_2$.[53] The oxidant most commonly used for this purpose is mercury(II) oxide.[54] Fluorenone hydrazone is converted in high yield into diazofluorene and many other diaryldiazomethanes can be prepared in the same way. Diazo ketones of relatively high stability, such as phenylbenzoyldiazomethane, can also be obtained by the oxidation of 1,2-diketone monohydrazones. Silver oxide reacts more rapidly with hydrazones than does mercury(II) oxide, and it is better for the preparation of monoaryldiazomethanes.[53] Activated manganese dioxide has also been used. The usual side products in these oxidations are azines $R^1R^2C{=}NN{=}CR^1R^2$, which are derived from the less stable diazo compounds by decomposition if the contact time with the oxidant is prolonged. Bishydrazones of 1,2-diketones are oxidized by mercury(II) oxide to alkynes (Scheme 14). This reaction provides a good method of synthesis of cycloalkynes such as cyclooctyne.[53]

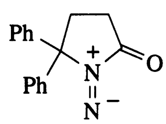


Scheme 14

The oxidation of 1,1-disubstituted hydrazines can be achieved by a wide range of oxidants. The oxidative removal of hydrogen formally leads to the production of aminonitrene, or 1,1-diazene, intermediates and many products of such reactions have been interpreted as being derived from aminonitrene intermediates.[55] Indeed, several of these species, derived from sterically hindered hydrazines by oxidation with nickel peroxide or *t*-butyl hypochlorite, have been detected and characterized in solution at low temperature. Examples include the diazenes **(14)**[56] and **(15)**[57]. The diazene **(16)** is stable enough to persist in solution at room temperature for several days.[58]

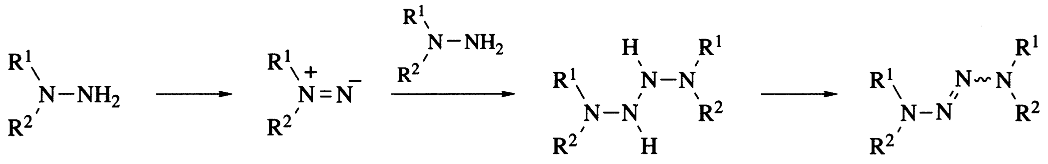


(14) **(15)**

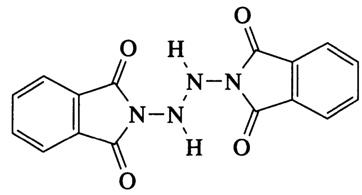


(16)

With many oxidants the most common products derived from 1,1-disubstituted hydrazines are the tetrazenes $R^1R^2NN{=}NNR^1R^2$. Benzeneselenic acid appears to a good reagent for their preparation[59] although several others, including mercury(II) oxide and nickel peroxide, have been widely used. These tetrazenes are formally the dimers of aminonitrenes and indeed are formed from long-lived species such as **(14)** by dimerization. Another possible mode of formation of tetrazenes, which is illustrated in Scheme 15, is the reaction of an aminonitrene, or its precursor, with the starting hydrazine to give a tetrazane, followed by further oxidation. This sequence was established for the oxidation of *N*-aminophthalimide by iodosylbenzene diacetate: the tetrazane (**17**) was isolated and gave the corresponding tetrazene with an excess of the oxidant.[60]

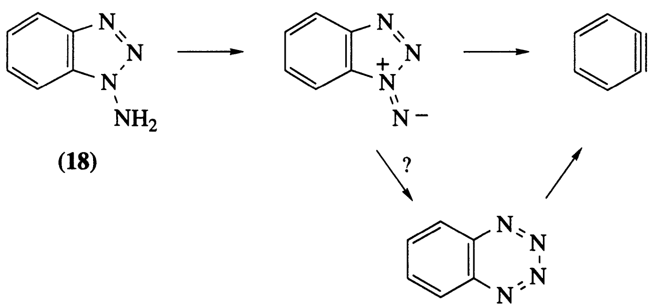


Scheme 15

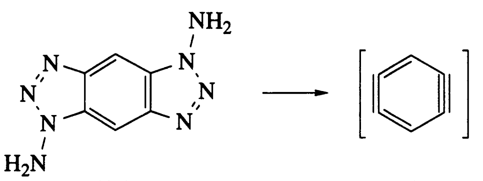


(17)

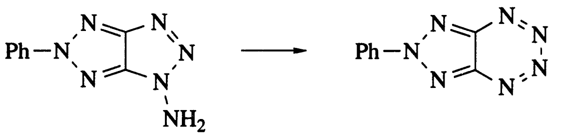
Lead(IV) acetate has proved to be a most efficient oxidant for these hydrazines. The oxidation of *N*-aminoheterocyclic compounds by lead(IV) acetate has, in particular, provided several very useful preparative procedures.[55,61] 1-Aminobenzotriazole (**18**; Scheme 16) is oxidized to 1,2-didehydrobenzene (benzyne) in high yield at low temperature.[62] This reaction has been widely exploited not only for the generation of benzyne but for the formation of cycloalkynes and of other arynes. For example, oxidation of the bisaminotriazole (**19**) gave products derived from 1,2,4,5-tetradehydrobenzene (**20**).[63] Campbell and Rees considered the possibility that benzyne was generated from 1-aminobenzotriazole by way of an aminonitrene and an unstable 1,2,3,4-benzotetrazine (Scheme 16).[62] A similar oxidation, of compound (**21**), did yield an isolable but unstable tetrazine (**22**) as the product.[64] 2-Aminobenzotriazole (**23**) clearly does not give the same intermediates as its isomer (**18**) on oxidation because it is oxidized cleanly to the dinitrile (**4**). On the other hand both 1- and 2-amino-3-phenylindazole give the same product of ring expansion, the benzotriazine (**24**), on oxidation (Scheme 17).[65]

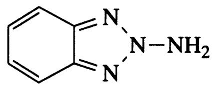


Scheme 16

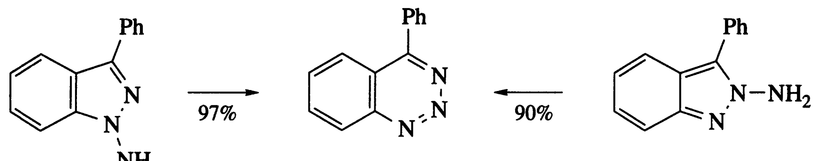


(19) **(20)**





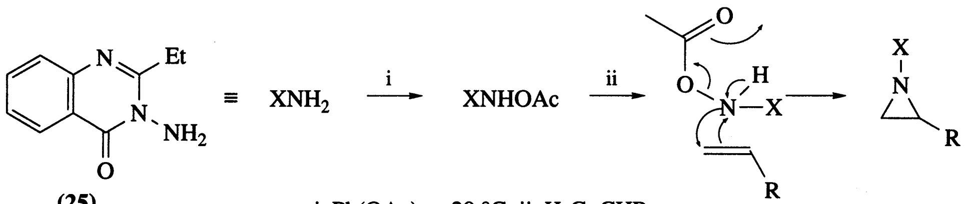
(21) **(22)** **(23)**



(24)

Scheme 17

The oxidation by lead(IV) acetate of *N*-aminophthalimide and of several *N*-aminolactams leads to the formation of intermediates which do not undergo fragmentation or rearrangement, but which can be intercepted by alkenes, alkynes, sulfoxides and other nucleophiles. The reactions have proved particularly useful for the synthesis of aziridines from a variety of alkenes.[55] The mechanism of these reactions has commonly been assumed to require the intermediacy of aminonitrenes, but this is probably not the case. Atkinson and Kelly have shown that oxidation of the aminolactam (**25**) by lead(IV) acetate at –20 °C leads to the formation of an unstable *N*-acetoxy compound.[66] This is the species which can form aziridines with alkenes. The mechanism shown in Scheme 18, which is analogous to that for the epoxidation of alkenes by peroxy acids, has been proposed for the aziridination process.



(25)

i, $Pb(OAc)_4$, –20 °C; ii, $H_2C{=}CHR$

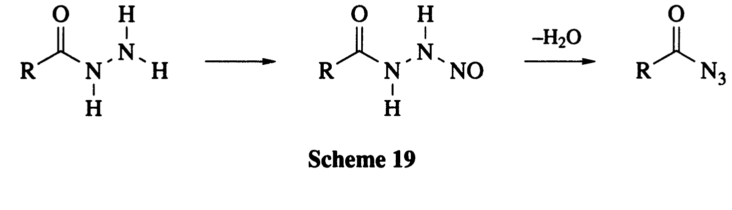
Scheme 18

Oxidation of methoxylamine and some other *O*-substituted hydroxylamines by lead tetraacetate in the presence of alkenes can also lead to the formation of aziridines. The oxidation of 2,4-dinitrobenzenesulfenamide is analogous.[67] In view of the results reported with aminolactams, these reactions do not necessarily establish the intermediacy of nitrenes in the oxidations.

6.1.2.2.2 Other oxidations

1,1-Diphenylhydrazine is oxidized to diphenylnitrosamine (50%) by potassium superoxide.[68] The same reagent also oxidizes 1-methyl-1-phenylhydrazine, but here the nitrosamine is a minor product; the major reaction is deamination. A better method of oxidative deamination of some 1,1-disubstituted hydrazines and hydrazinium salts is reaction with nitrous acid. Thus, several hydrazinium salts $Me_2RN^+NH_2\ X^-$ were deaminated to the tertiary amine by treatment with nitrous acid.[69] The method has also been used to deaminate *N*-aminoheterocyclic compounds; for example, some 1,2,3-triazoles are conveniently prepared by deamination of the corresponding 1-aminotriazoles with nitrous acid.[70]

Monosubstituted hydrazines and hydrazides are converted into azides by a variety of nitrosating agents. The mildest reagent appears to be dinitrogen tetroxide, which can be used below 0 °C in acetonitrile to convert benzoylhydrazine, *p*-toluenesulfonylhydrazine and 4-nitrobenzoylhydrazine, among others, into the corresponding azides in high yield.[71] Another mild method involves the use of iron(III) nitrate supported on clay.[72] These reactions probably proceed by way of transient *N*-nitroso compounds (Scheme 19).



Scheme 19

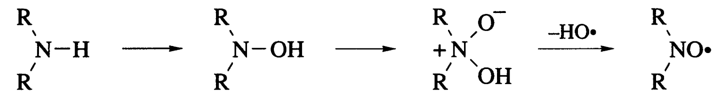
Katritzky and coworkers have nitrated hydrazinium salts and 1,1-disubstituted hydrazines with nitronium tetrafluoroborate and other nitrating agents.[73] Thus, nitrimides $R_3N^+N^-NO_2$ were prepared in good yield from hydrazinium salts derived from tertiary amines such as trimethylamine and 2-methylpyridine; 1-aminobenzotriazole was also nitrated at the amino group.

6.1.3 OXIDATION OF THE NH GROUP

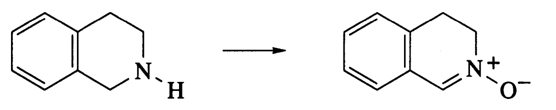
6.1.3.1 Secondary Amines

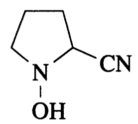
6.1.3.1.1 Oxidation to nitroxides and hydroxylamines

The conversion of secondary amines R_2NH into nitroxides $R_2NO\cdot$[74] has been carried out using hydrogen peroxide, MCPBA and metal oxides, including silver oxide, mercury(II) oxide and lead(IV) oxide. Secondary amines which have low solubility in water have been oxidized by sodium tungstate and hydrogen peroxide in a mixture of methanol and acetonitrile.[5] Dimethyldioxirane (**1**) is also a good reagent for the oxidation of hindered secondary amines to nitroxides.[75] As illustrated in Scheme 20, such oxidations probably go by way of the hydroxylamines as intermediates. When the secondary amine has a hydrogen atom attached to an α-carbon atom, the hydroxylamine formed by oxidation with sodium tungstate and hydrogen peroxide reacts further by losing this hydrogen atom. For example, tetrahydroisoquinoline was oxidized to the nitrone (**26**; 85%).[76] The reaction has been modified to provide a method of synthesis of *N*-hydroxy amino acids from secondary amines. Thus, *N*-hydroxyproline was prepared from pyrrolidine by oxidation with hydrogen peroxide and sodium tungstate, the intermediate being intercepted by cyanide. This gave the hydroxylamine (**27**) which could then be converted into *N*-hydroxyproline by hydrolysis.[77] A related oxidation has been used for the conversion of tetrahydroquinolines into the corresponding 3,4-dihydro-1-hydroxy-2-quinolones.[78]



Scheme 20



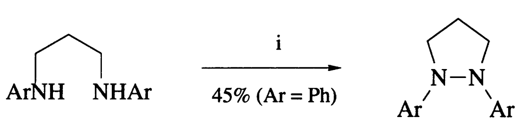


(26) **(27)**

Dibenzoyl peroxide oxidizes morpholine, piperidine and other simple secondary amines in good yield to the corresponding benzoyloxyamines; these compounds can then be hydrolyzed in basic conditions to the free hydroxylamines.[79] An analogous reaction takes place between secondary amines and bis(diphenylphosphinyl) peroxide; for example, diethylamine is converted into the hydroxylamine derivative $Et_2NOPOPh_2$ (97%) by this reagent.[80] The products are easily hydrolyzed to the free hydroxylamines, and they can also be used as aminating agents.

6.1.3.1.2 Oxidation to aminium ions, aminyl radicals and hydrazines

The one-electron oxidation of a secondary amine results in the formation of a secondary aminium ion[1] which on deprotonation gives an aminyl radical (Scheme 1).[2] The nature of the final products derived from these intermediates depends very much on the structure of the substrate and the reaction conditions. If the amine has a hydrogen atom on the α-carbon atom the major products usually result from deprotonation at this α-position. With aromatic secondary amines, products can result from coupling of the delocalized radicals at a ring carbon atom. The formal dimerization of aminyl radicals shown in Scheme 21 is therefore not often a useful method of preparation of hydrazines. Nickel peroxide has been used to oxidize diphenylamine to tetraphenylhydrazine in moderate yield, and other secondary arylamines also give mixtures of products in which the corresponding hydrazines are the major components.[25] The reagent formed by the addition of oxygen to copper(I) chloride in pyridine can also oxidize secondary arylamines. Diphenylamine was oxidized to tetraphenylhydrazine in 83% yield, and *N*-methylaniline gave the corresponding hydrazine in 52% yield.[81] Lithium piperidide and lithium salts of other secondary amines have also been oxidized to the hydrazines with copper(I) chloride and oxygen.[82] Intramolecular coupling of the diamines (**28**) to the pyrazolines (**29**) has also been achieved, using activated manganese dioxide (Scheme 21).[5,83]



(28) **(29)**

i, MnO_2, −20 °C

Scheme 21

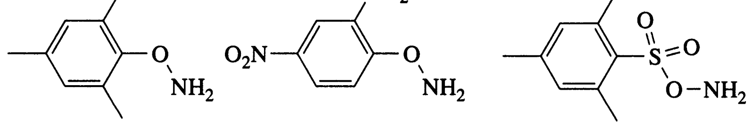
6.1.3.1.3 Nitrosation and nitration

Methods of formation of *N*-nitrosamines from secondary amines have been reviewed.[84] The most widely used reagent is sodium nitrite in an aqueous acidic medium; others include nitrosyl chloride, nitrogen oxides and nitrite esters. Fremy's salt [$2K^+$ $(SO_3^-)_2NO\cdot$] reacts with hydroxylamine in the presence of secondary amines to give *N*-nitrosamines, or, in the presence of an excess of hydroxylamine, tetrazenes $R_2NN{=}NNR_2$. The nitrosating agent derived from hydroxylamine and Fremy's salt is suggested to be the anion $(SO_3^-)_2NONO$.[85]

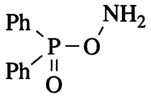
Direct *N*-nitration of secondary amines by nitric acid is possible only for weakly basic amines.[86] The more basic amines can be nitrated under neutral conditions with reagents such as dinitrogen pentoxide and nitronium tetrafluoroborate, but nitrosamines are significant by-products. The nitrate ester $CF_3CMe_2ONO_2$ has been recommended as a nonacidic nitrating agent for secondary amines which avoids the problem of contamination of the products by *N*-nitrosamines: piperidine and pyrrolidine were nitrated in yields of 75% and 72%, respectively.[87] Amides and imides are efficiently *N*-nitrated using ammonium nitrate in trifluoroacetic anhydride.[88]

6.1.3.1.4 Amination

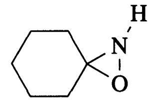
The conversion of secondary amines into 1,1-disubstituted hydrazines requires the use of an electrophilic aminating agent. Several such reagents are available, and their use has been reviewed.[46,89–91] Chloramine and hydroxylamine-*O*-sulfonic acid are the commonest reagents of this type. Secondary amines have been aminated by chloramine in moderate yield in aqueous solution, and in good yield by passing gaseous chloramine into methanolic solutions of the amines.[46] Hydroxylamine-*O*-sulfonic acid is a more powerful aminating agent and has been used to aminate heterocyclic compounds such as benzotriazole in good yield. Its disadvantage is its low solubility in nonpolar organic solvents. To overcome this problem several soluble *O*-substituted hydroxylamines have been introduced as aminating agents. These include *O*-mesitylhydroxylamine (**30**), *O*-2,4-dinitrophenylhydroxylamine (**31**), and *O*-mesitylenesulfonylhydroxylamine (MSH; **32**). MSH is a particularly good aminating agent, but it is rather tedious to prepare.[90] A promising, and much more accessible, reagent is *O*-diphenylphosphinylhydroxylamine (**33**).[92] This reagent has been used to aminate imides such as phthalimide in high yield. The oxaziridine (**34**) can act as an aminating agent for amino acids and peptides.[93]



(30) **(31)** **(32)**



(33)



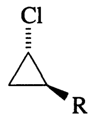
(34)

6.1.3.1.5 Halogenation and other reactions

Methods for the chlorination of secondary amines, secondary amides and imides have been reviewed.[48] Secondary alkylamines can be chlorinated by *t*-butyl hypochlorite at low temperature, or by sodium hypochlorite or *N*-chlorosuccinimide. *N*-Bromination and *N*-iodination can be brought about by using the appropriate halogen, but these *N*-halodialkylamines are unstable and are rarely isolated. *N*-Fluoroamines and *N*-fluoroamides are also rare.[94] The *N*-fluoroimide $(CF_3SO_2)_2NF$, has, however, been prepared by direct fluorination with fluorine. It is a stable liquid which shows promise as a fluorinating agent for aromatic compounds.[95]

N-Chloroaziridines (**35**) are configurationally stable, and there have been attempts to prepare them in optically active form by carrying out the chlorination with *t*-butyl hypochlorite or with *N*-chlorosuccinimide in the presence of the optically active alcohol (*S*)-(+)-$PhCH(OH)CF_3$.[96] Optical yields were, however, low (less than 10% *ee*).

Secondary amines can be converted into sulfenamides by reaction of the lithium dialkylamides with disulfides.[50,97] Perchlorylamines (**36**) have been prepared in good yields from piperidine and other secondary amines by reaction with chlorine(VII) oxide.[98]



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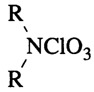
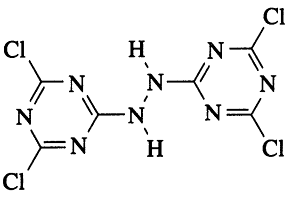


(36)

6.1.3.2 Hydrazines and Hydroxylamines

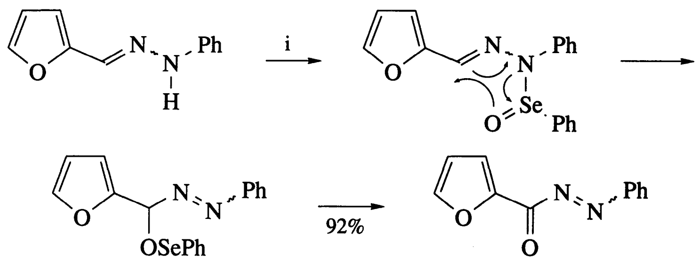
Methods of oxidation of hydrazo to azo compounds[99] and hydroxylamino to nitroso compounds[100] have been reviewed. Reagents which oxidize aromatic primary amines to azo compounds are also suitable for the oxidation of aromatic hydrazo compounds, since the hydrazo compounds are intermediates in the oxidation of the amines. Thus, manganese dioxide, mercury(II) oxide and lead tetraacetate are all suitable oxidants. Silver carbonate on Celite rapidly oxidizes both diarylhydrazines and acylhydrazines to the corresponding azo compounds in good yield.[26] Another supported oxidant which can convert hydrazobenzene into azobenzene in high yield is sodium periodate on silica gel.[101]

Two-phase oxidation systems are useful for the selective oxidation of the hydrazo to the azo group in the presence of other functional groups. For example, the hydrazotriazine (**37**) was oxidized by chlorine to the azo compound in high yield, without hydrolysis of the delicate ring chloride, in a two-phase system of chloroform and aqueous sodium bicarbonate.[102] Aqueous potassium ferricyanide has also been used in a two-phase system to oxidize hydrazo compounds. A coreagent is required, which is either a hindered phenol[103] or carbon black.[104] In either case the active oxidant is believed to be an aryloxy radical. This type of reagent has been used to produce AIBN and acylazo compounds such as PhN=NCOPh from the corresponding hydrazo compounds in high yield. Oxygen can also oxidize such hydrazo compounds in the presence of a palladium catalyst at room temperature.[105] Benzeneseleninic anhydride has been used to convert arylhydrazones of aldehydes into α-carbonylazo compounds: for example, furfural phenylhydrazone gave the azo compound (**38**) in 92% yield (Scheme 22).[106] Arylhydrazones



(37)

$R_2C{=}NNHAr$ are also oxidized to α-acetoxyalkylazo compounds $R_2C(OAc)N{=}NAr$ by lead tetra-acetate.[99]

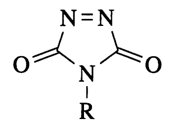


(38)

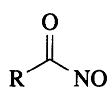
i, $(PhSeO)_2O$

Scheme 22

Several oxidants have been used to produce the highly reactive cyclic azocarbonyl compounds (**39**) from the corresponding hydrazides.[107] These include *t*-butyl hypochlorite, dinitrogen tetroxide and *N*-bromosuccinimide. Anodic oxidation[108] and oxidation by iodosylbenzene diacetate[109] are also effective for preparing these and related azocarbonyl compounds.



(**39**)



(**40**)

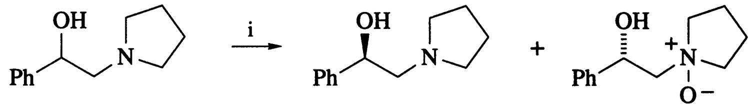
Aliphatic nitroso compounds can be prepared from *N*-alkylhydroxylamines by oxidation with bromine, chlorine or sodium hypochlorite in weakly acidic solution, by reaction with potassium dichromate in acetic or sulfuric acid, and by oxidation with yellow mercury(II) oxide in suspension in an organic solvent.[100] Silver carbonate on Celite has also been used to prepare aliphatic nitroso compounds, such as nitrosocyclohexane, in high yield from the corresponding hydroxylamines.[110] Aqueous sodium periodate and tetraalkylammonium periodates, which are soluble in organic solvents, are the reagents most commonly used for the oxidation of hydroxamic acids and *N*-acylhydroxylamines to acylnitroso compounds (**40**).[111] These compounds are rarely isolated, but are useful as highly reactive dienophiles in the Diels–Alder reaction.[112]

6.1.4 OXIDATION OF TERTIARY (sp^3) NITROGEN

6.1.4.1 Formation of *N*-Oxides

Tertiary alkylamines can be converted into the corresponding *N*-oxides with hydrogen peroxide or with peroxy acids.[113] *t*-Butyl hydroperoxide has also been used in the presence of a catalyst such as $VO(acac)_2$. Sharpless and coworkers have carried out the oxidative kinetic resolution of several β-hydroxy tertiary amines such as (**41**) with *t*-butyl hydroperoxide, titanium(IV) isopropoxide and (+)-diisopropyl tartrate, the titanium(IV):tartrate ratio being about 2:1.[114] After 60% conversion, one enantiomer was selectively oxidized, and the other enantiomer could be recovered in good optical purity (Scheme 23).

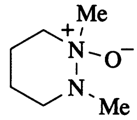
Tertiary alkylamines are normally autoxidized and dealkylated by oxygen, but it has been shown that they can slowly be converted into the *N*-oxides by heating at 90–130 °C with oxygen under pressure in a polar solvent.[115] The reaction is thought to involve electron transfer from the amine to oxygen as the rate-determining step. Pyridine does not react with oxygen under these conditions. The cyclic hydrazine *N*-oxide (**42**) has been prepared from the corresponding hydrazine by oxidation with 30% hydrogen peroxide at room temperature.[116]



(41) 57%, 95% *ee* 59%, 63% *ee*

i, Bu^tOOH, $Ti(OPr^i)_4$, (+)-diisopropyl tartrate; 60% conversion

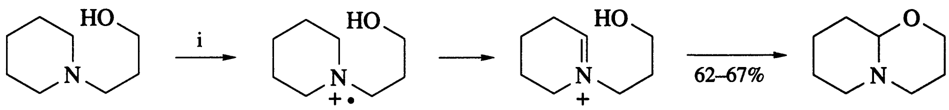
Scheme 23



(42)

6.1.4.2 Other Reactions

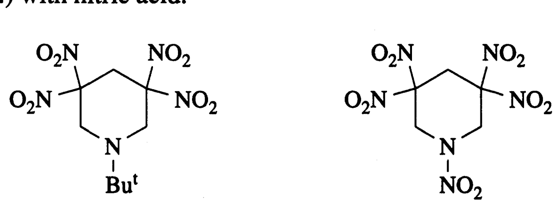
One-electron oxidants convert tertiary amines into aminium ions.[1] Chlorine dioxide[5] is an example of such a reagent; this converts triethylamine and other tertiary alkylamines into aminium ions, which normally react further by the loss of hydrogen from an α-carbon atom. An example of a synthetic application of this reaction sequence is shown in Scheme 24, the intermediate iminium ion being intercepted intramolecularly.[117] Nitrosonium tetrafluoroborate and dioxygenyl hexafluoroantimonate, $O_2{}^+SbF_6{}^-$, are also good one-electron oxidants which can be used at low temperature.[118]



i, ClO_2, pH 9–11

Scheme 24

Reagents for amination, nitrosation and nitration of tertiary alkylamines are discussed in the appropriate reviews listed in Sections 6.1.3.1.4 and 6.1.3.1.5. Tertiary amines can be nitrosated with dealkylation by dinitrogen tetroxide: for example, 1-methylpiperidine gave 1-nitrosopiperidine (80%).[119] This reaction probably starts by one-electron oxidation of the amine, the aminium ion then undergoing dealkylation. Other oxidative deacylations and dealkylations include the formation of *N*-nitrosodibenzylamine in high yield from the acid chloride $(PhCH_2)_2NCOCl$ and sodium nitrite[120] and the conversion of the amine (**43**) into the nitramine (**44**) with nitric acid.[121]

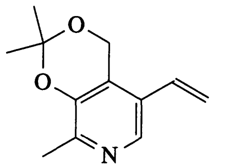


(43) **(44)**

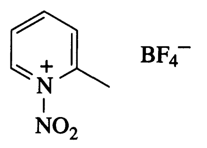
6.1.5 OXIDATION OF TRIGONAL (sp^2) NITROGEN

6.1.5.1 *N*-Oxidation of Heteroaromatic Amines

The standard methods of oxidation of pyridines and other heteroaromatic nitrogen compounds make use of peroxy acids or hydrogen peroxide in carboxylic acid solution.[122] Peracetic acid, peroxymonophthalic acid and MCPBA can all convert simple pyridines to the *N*-oxides. For example, the pyridine (**45**) was oxidized at nitrogen by MCPBA without attack on the vinyl group.[123] Peroxymaleic acid has been used to convert deactivated heteroaromatics, such as 2-chloroquinoline, into the *N*-oxides.[122] Trifluoroperacetic acid, or hydrogen peroxide in trifluoroacetic acid can also oxidize deactivated and hindered heteroaromatic compounds; for example, 2,6-dibromopyridine was oxidized in good yield by 30% hydrogen peroxide in trifluoroacetic acid.



(45)



(46)

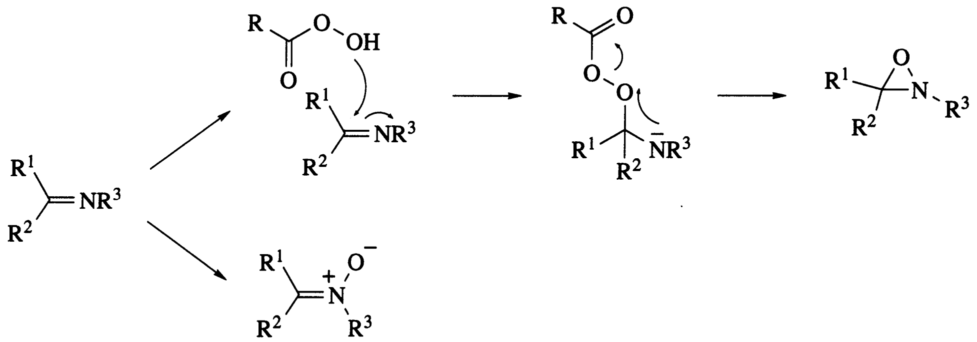
These and more powerful oxidants are often required for the oxidation of diazines and triazines.[124] Chivers and Suschitzky succeeded in oxidizing pentafluoropyridine, tetrachloropyrazine and other polyhalogenated azines with 90% hydrogen peroxide in acetic and sulfuric acids or in trifluoroacetic and sulfuric acids.[125] The hazards of handling 90% hydrogen peroxide solutions can be avoided if the commercially available solid adduct formed between urea and hydrogen peroxide is used instead.[126] Diazines and diazanaphthalenes which are sensitive to peroxy acids can be oxidized by hydrogen peroxide in the presence of a sodium tungstate catalyst.[127] Sodium perborate in acetic acid is an especially useful oxidant for water-soluble pyrazine *N*-oxides.[128]

Dimethyldioxirane (**1**) is a mild and efficient oxidant for pyridine; pyridines and quinolines have also been oxidized in good yield by *t*-pentylhydroperoxide in the presence of molybdenum(V) chloride.[129]

N-Halogenation and other oxidative reactions of pyridines and related heterocycles have been reviewed.[130] Thus, pyridines and some diazines[124] can be aminated by mesitylenesulfonylhydroxylamine (**32**) and similar reagents. *N*-Nitration of 2-picoline by nitronium tetrafluoroborate gives the salt (**46**) which is itself a nitrating agent.

6.1.5.2 *N*-Oxidation of Imines

Imines are oxidized by peroxy acids to oxaziridines, with the formation of nitrones as a competing process (Scheme 25).[131] Oxaziridines are probably formed by a two-step process involving nucleophilic addition of the peroxy acid to the C═N bond, followed by elimination, as illustrated; nitrones are formed by competing *N*-oxidation.[4] The oxidation of chiral imines to oxaziridines by MCPBA proceeds with good, but not complete, diastereoselectivity.[132]

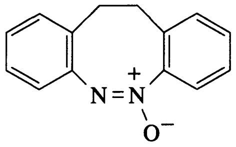


Scheme 25

6.1.5.3 Oxidation of Azo to Azoxy Compounds

Methods for the oxidation of azo to azoxy compounds have been reviewed.[133] Typical oxidants are MCPBA and other peroxy acids and hydrogen peroxide. *t*-Butyl hydroperoxide is also an effective oxidant for azobenzenes in the presence of molybdenum hexacarbonyl.[134] Trifluoroperacetic acid has been used to oxidize perfluoroazobenzene and other fluorinated azobenzenes to the azoxy compounds in high yield;[135] It is also capable of oxidizing azoxyarenes, such as compound (**47**), to the di-*N*-oxides.[136]

Bridgehead azoalkenes (**48**) can be oxidized to the azoxy compounds and to the di-*N*-oxides with MCPBA.[137]



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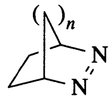
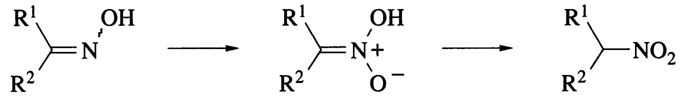


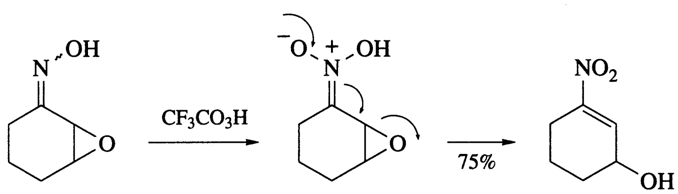
(48)

6.1.5.4 Oxidation of Oximes and Nitroso Compounds

The *N*-oxidation of oximes leads to the formation of *aci*-nitro compounds. In a polar organic solvent such as acetonitrile these compounds are isomerized to nitro compounds (Scheme 26) and are thus protected from further oxidation. The reagent of choice is trifluoroperacetic acid.[138] The oxidation of oximes derived from α-epoxy ketones results in the formation of a nitroalkene with opening of the epoxide, as illustrated in Scheme 27.[139]

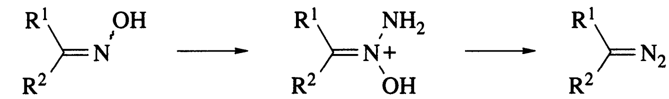


Scheme 26

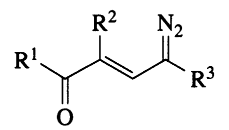


Scheme 27

N-Nitrosation of oximes by nitrosyl halides or nitrite esters often results in the formation of *N*-nitrimines, $R^1R^2C{=}NNO_2$, these compounds being formed by rearrangement of the initial adducts.[140] *N*-Amination of oximes by chloramine or by hydroxylamine-*O*-sulfonic acid can result in the formation of diazo compounds (Scheme 28). The reaction, known as the Forster reaction,[53] has been used for the preparation of aryldiazoalkanes, although better methods are usually available. Diazo ketones of the general formula (**49**) have also been prepared by this method from the oximes.[141]



Scheme 28



(49)

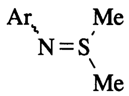
The oxidation of oximes by lead(IV) acetate, chlorine and other oxidants results in the formation of α-substituted nitroso compounds $R^1R^2C(X)NO$ by attack of the oxidant at carbon. This is also the reaction commonly observed with dinitrogen tetroxide (the Ponzio reaction), the products being *gem*-

dinitro compounds or α-nitronitroso compounds.[142] It has, however, been claimed that many alkyl ketoximes can be converted into the corresponding ketones in good yields by reaction with dinitrogen tetroxide at low temperature.[143] Other oxidative methods of deprotection of oximes exist.[143,144]

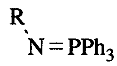
Nitroso compounds are readily oxidized to the corresponding nitro compounds by peroxides, peroxy acids, oxygen, ozone and dinitrogen tetroxide.[4]

6.1.5.5 Other Reactions

Reactions which formally involve the oxidation of azides have been reviewed by Boyer.[4] Other oxidations with useful synthetic applications include two which start from nitrogen ylides. Sulfimides (**50**) derived from electron-deficient aromatic and heterocyclic amines are oxidized to the corresponding nitroso compounds by MCPBA.[4,145] This is a very useful method of preparation of some otherwise inaccessible nitroso compounds such as 2-nitrosopyridine and 1-nitrosoisoquinoline. They can be further oxidized, for example by ozone, to the nitro compounds. Phosphimides (**51**) are oxidized directly by ozone to the nitro compounds, although the nitroso compounds are intermediates.[146] Isocyanates can also be oxidized to the corresponding nitro compounds, by dimethyldioxirane (**1**).[147]



(50)

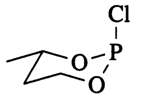


(51)

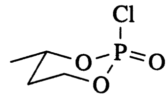
6.1.6 OXIDATION OF PHOSPHORUS

A comprehensive survey of the chemistry of organophosphorus compounds was published in 1982.[148] This provides descriptions of many of the oxidation methods in phosphorus chemistry, and primary literature references to them. Primary references are not given here if they are available from this source.

Tertiary phosphines (phosphanes) are normally very easily oxidized to the corresponding phosphine oxides.[149] The reaction can be brought about by oxygen alone or with hydrogen peroxide. Bis(trimethylsilyl) peroxide has been recommended as a selective oxygenating agent for phosphines and phosphites.[150] Oxidation of the chiral phosphine (*R*)-(–)-methylpropylphenylphosphine proceeded with 95% retention of configuration and the chlorodioxaphosphorinane (**52**) gave the oxide (**53**) with complete retention of stereochemistry. Diaryl selenoxides[151] and sulfur trioxide[152] have also been used for the oxidation of phosphines. Electrochemical oxidation of phosphines and phosphites proceeds in the presence of diethyl disulfide, which is effectively a catalyst for the reaction.[153]



(52)

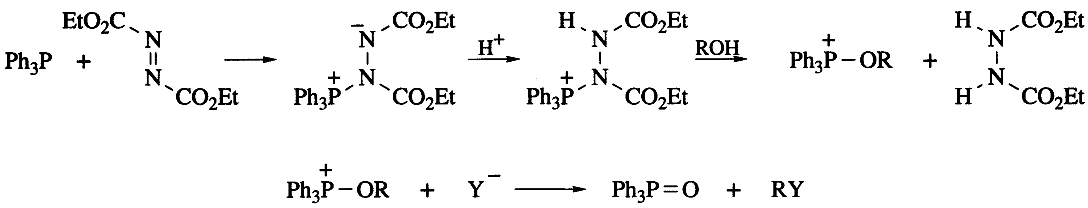


(53)

Tertiary phosphine sulfides and selenides are readily obtained from the phosphines and elementary sulfur or selenium. Potassium selenocyanate provides a convenient alternative to selenium for preparing selenides.[149] Phosphinimides (iminophosphoranes), $R_3P{=}NR'$, are prepared by the reaction of phosphines with azides or with *N*-chloroamines, among other methods. Phosphines can be aminated by reaction with *O*-diphenylphosphinylhydroxylamine.[92] The reaction of triphenylphosphine with diethyl azodicarboxylate results in rapid conjugate addition of the phosphine to the N=N bond. The resulting betaine has been used as a method of activation of alcohols (the Mitsunobu reaction), allowing displacement by nucleophiles and conversion of the alcohols into esters and other derivatives (Scheme 29).[154]

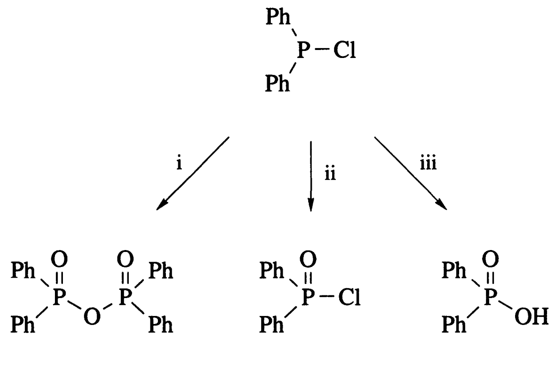
Many reagents, including the halogens themselves, are suitable for the conversion of trisubstituted phosphines into dihalides R_3PX_2 (X = F, Cl, Br or I).[155] These compounds can exist either in the pentacovalent form, or in an ionic form R_3PX^+ $R_3PX_3^-$, the covalent structure being favored in the order F > Cl > Br > I and by solvents of low polarity.

Secondary phosphines, R_2PH, can be chlorinated under controlled conditions to give the corresponding chlorophosphines.[156] These compounds can be further oxidized in a number of ways (Scheme 30).[157] The reaction of chlorodiphenylphosphine with dibenzoyl peroxide gives diphenylphosphinic acid



Scheme 29

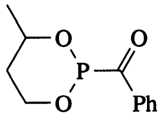
anhydride in 82% yield. With sulfur dioxide, diphenylphosphinyl chloride is formed in high yield. Oxidation with oxygen, or with sodium hydride and oxygen in the presence of a secondary alcohol such as diphenylmethanol, gives diphenylphosphinic acid after hydrolysis. Diphenylphosphinic anhydride can also be obtained by oxidation of the bisoxide $(Ph_2PO)_2$ with perbenzoic acid.



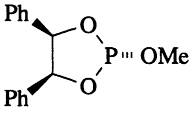
i, $(PhCO_2)_2$; ii, SO_2; iii, NaH, Ph_2CHOH, O_2

Scheme 30

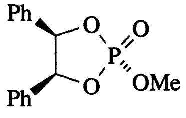
Derivatives of phosphonic acids, $RP{=}O(OH)_2$, can be prepared by several different oxidative methods.[158] Primary phosphines RPH_2 are oxidized to phosphonic acids by hydrogen peroxide or by sulfur dioxide: thus, phenylphosphine gave benzenephosphonic acid (96%) on reaction with sulfur dioxide at room temperature in a sealed tube. Phosphinic acids, $RP{=}O(OH)H$, can also be oxidized to the corresponding phosphonic acids with hydrogen peroxide. Ozone oxidized the dioxaphosphorane (**54**) to the phosphonic ester in 73% yield. Ozone is also capable of stereospecific oxidation of phosphite esters to phosphates.[159] For example, the cyclic phosphite (**55**) was oxidized to the phosphate (**56**) with retention of configuration. Peroxy acids and selenium dioxide are other common oxidants for phosphite esters.[160]



(54)



(55)



(56)

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